Sahaj Bhadja

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Summary

Junior software engineer with 2+ years implementing and supporting **object-oriented applications**, relational databases, and internal workflows; experienced in coding, debugging, testing, performance tuning, automation, documentation, SQL Server, Oracle, and full lifecycle delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Implemented and maintained **Python applications and relational database structures** for 500+ users across SQL Server and PostgreSQL, tuning indexed queries and access patterns to reduce latency from 1.2 seconds to 350 milliseconds.
- Supported **production software functionality** across 5,000+ monthly jobs, debugging user-reported integration failures and adding validation, reconciliation, error handling, and documentation to achieve 99%+ reliability through tested CI/CD releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Delivered **full lifecycle React and Flask application features** for uploads, processing, history, and reporting, supporting 150+ end users and 10,000+ daily requests through maintainable MVC architecture and REST services.
- Handled **application debugging, testing, and production support** by reproducing file-processing defects, adding secure validation, persistent results, fallback handling, and automated deployments, reducing turnaround time 60% under peak usage.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Refactored **object-oriented API and database components** using Singleton and Factory patterns, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Performed **software testing, quality assurance, and release automation** with Jenkins, PyTest, Mockito, and 20+ unit/integration tests, generating coverage and failure reports across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed and supported **React, Node.js, and Azure SQL business applications** across eight services for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs and WebSockets.
- Improved **performance, usability, and operational reliability** through Jest, Selenium, Sentry, Azure Monitor, Docker, and Jenkins CI/CD, debugging software defects and cutting deployment time 50% across internal production workflows.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Spring Boot and Flask application services** for bookings, pricing, payments, and status updates, integrating Firebase persistence and validated webhooks while maintaining sub-200-millisecond responses across 50,000+ relational-style records.
- Collaborated with users to **document requirements and maintain functionality**, mapping 12+ integrations and delivering unit-tested, version-controlled enhancements that improved cross-platform compatibility 15%, reduced design errors 25%, and accelerated delivery 22%.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
Applications: Spring Boot, FastAPI, Flask, Node.js, React.js, MVC, REST APIs, WebSockets, microservices
Databases: Oracle, SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, Firebase, relational design, indexing
Quality: Software testing, quality assurance, debugging, performance tuning, automation, PyTest, Mockito, Jest, Selenium
Delivery: OOP, documentation, end-user support, Git, Jenkins, GitHub Actions, CI/CD, Docker, Agile/Scrum

Projects

Chat2SQL Code | *Java, TypeScript, Oracle, Docker, Maven*
- Enhanced a **cross-platform database application** across Java and TypeScript components, supporting Oracle and 10+ SQL/NoSQL systems with schema exploration, query generation, data management, Docker, Maven, and structured debugging workflows.

HMIS Live | *MERN, Microservices, OOP*
- Developed and supported a **full-stack operations application** with role-based workflows, REST integrations, TypeScript dashboards, notifications, and Sentry diagnostics, halving record retrieval time while reducing administrative workload 30%.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Deployed a **containerized Python application** to AWS ECS/Fargate using CodePipeline, automated validation, and versioned CI/CD, achieving 87% accuracy and reliable, repeatable builds, installation, testing, maintenance, and releases.